

Computer Practical: Parcel Modelling

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1 Overview

In this computer practical, you will use a parcel model to investigate how water vapour condenses onto aerosol particles to form cloud droplets. As in previous practicals, the model advances a set of coupled ordinary differential equations in time.

By the end of the practical you should be able to:

- explain why a rising air parcel cools and eventually forms cloud;
- describe how vertical velocity and initial humidity affect cloud formation;
- explain how aerosol number, size-distribution width and particle size affect cloud droplet number
- configure, run and analyse a compiled environmental model.

2 Model formulation

This model calculates the properties of an air parcel as it moves vertically through the atmosphere. Atmospheric pressure changes with height; expansion associated with decreasing pressure cools a rising parcel and can eventually lead to condensation. The model solves several coupled equations as below:

$$e_0 = 610.7 \exp \left(\frac{L_v}{R_v} \left[\frac{1}{273.15} - \frac{1}{T} \right] \right) \quad (1)$$

$$e_v = e_0 \exp \left(\frac{2\sigma_w}{R_v T \rho_l r_i} \right) \quad (2)$$

$$\frac{dP}{dt} = -\rho_a g w \quad (3)$$

$$\frac{dT}{dt} = \frac{R_a}{c_p} \frac{T}{P} \frac{dP}{dt} + \frac{L_v}{c_p} \frac{dw_l}{dt} \quad (4)$$

$$\frac{dm_i}{dt} = 4\pi r_i D_v \left(w_{v,\infty} \rho_a - \frac{e_v(T)}{R_v T} \right) \quad (5)$$

$$\frac{dw_l}{dt} = \sum_i^N N_i \frac{dm_i}{dt} \quad (6)$$

$$\frac{dw_{v,\infty}}{dt} = -\frac{dw_l}{dt} \quad (7)$$

where e_0, e_v are the equilibrium vapour pressure over a flat and curved surface respectively; L_v is the latent heat of vaporisation, R_v, R_a are the specific gas constants for water vapour and air respectively; T is temperature; σ_w is the surface tension of water; ρ_l, ρ_a are the densities of liquid water and air respectively; r_i is the radius of an aerosol particle / droplet; P is pressure; g is gravity; c_p is the specific heat capacity of air; m_i is the mass of a single aerosol particle / drop; $w_{v,\infty}$ is the mass of water vapour in the parcel; N_i is the number of aerosol particles / droplets in the parcel; w is the vertical wind, and D_v is a constant.

These are the equations that the FORTRAN model will solve for us.

3 Aerosol size distributions

In the atmosphere, aerosol particle size distributions tend to be distributed according to a log-normal distribution. There are 3 parameters that describe the shape of a log-normal distribution: N , $\bar{D}$, σ_g and we will use these variables as part of our initial conditions. In the atmosphere, aerosol particles tend to come in 3 of these modes:

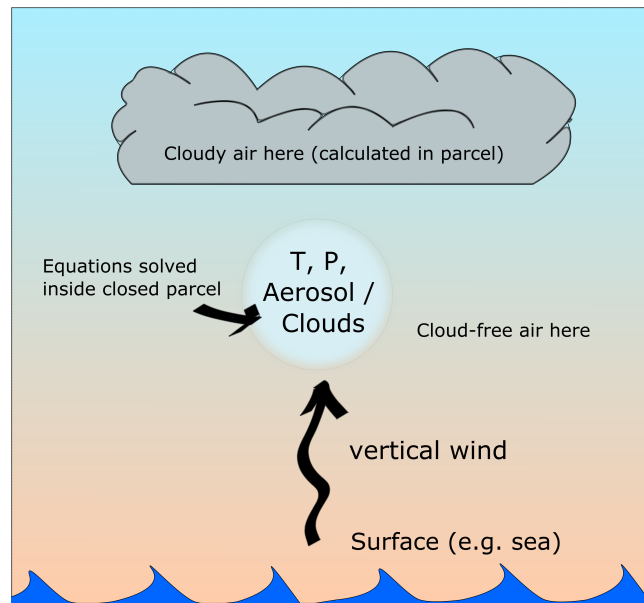


Figure 1: Schematic of the scenario being modelled. The model variables are within the small parcel of air shown. However, if we assume the air surrounding the parcel has similar properties to the parcel we can use the model to predict the properties of the atmosphere along a vertical coordinate.

the nucleation mode, the accumulation mode, and the coarse mode, so there will be 3 modes in the input file and water will condense onto these aerosol particles to form a cloud. However, the idea with the Marine Cloud Brightening geoengineering scheme is that adding sea salt aerosol particles to the background aerosol will alter the properties of the clouds making them more reflective. This is the idea we will explore in the practical today.

The parcel model is written in Fortran and solves for the movement of an adiabatic parcel of air along a vertical coordinate in the atmosphere. The model generates an output file in a format known as NetCDF. NetCDF is a common data format used in atmospheric and environmental sciences. This file can then be read in and plotted using a Python script. Figure 1 illustrates the main idea behind the model simulations that you will run today.

4 Downloading and compiling the code

Log into the server computer with an SSH window. Also, log into the server computer with an SFTP window. Space the windows out on your screen so that you can easily move between them.

If you have not done this already, download the code you will be using today from GitHub by typing the following:

```
git clone https://github.com/EnvModelling/bin-microphysics-model
```

this should download the code to your working directory.

In order to access the files you will need to change the working directory to where the files are. Type the following:

```
cd bin-
```

followed by the `tab` key and the command should auto-complete. Then press the `enter` key.

Type `ls` followed by the `enter` key. The screen should list the files in this directory.

Finally, type `make` followed by the `enter` key. This will compile the FORTRAN code into machine code that the computer can run.

Table 1: Parameters that you will change, which control the behaviour of the model.

Variable	Default value	Description
winit	0.3	the speed that the parcel moves along the vertical in the atmosphere (m s^{-1})
tinit	290	the initial temperature in kelvin.
pinit	100000	the initial pressure (Pa).
rhinit	0.65	the initial relative humidity (fraction).
n_aer1	46.64e6, 153.42e6, 166.77e6, 0e6, 0e6, 0.e6	number concentration in each mode (kg^{-1}).
d_aer1	18e-9, 39e-9, 154.e-9, 1e-6, 1e-6, 100.e-9	number-median dry diameter in each mode (m).
sig_aer1	0.348, 0.354, 0.465, 0.3, 0.3, 0.5	$\ln \sigma_g$ for each mode (width).

5 Viewing and Editing the code

We will use the `nano` text editor to view some files. The file that takes input data is called `namelist.in`. To see what variables we can change, type:

```
nano -l namelist.in
```

the `-l` means to show line numbers. The file can be edited in the text editor and saved by typing `Ctrl-X`, then `Y`, followed by `Enter`.

Rather than relying only on line numbers, you can search for a parameter in `nano` using `Ctrl-W`. For example, search for `winit`, `rhinit` or `n_aer1`. In the current version of the file, `winit` is on line 18, `rhinit` is on line 28, and the aerosol size-distribution parameters are on lines 161–168.

The arrays `n_aer1`, `d_aer1` and `sig_aer1` have 3×2 elements: three log-normal submodes for each of two externally mixed aerosol types. The first three values describe the background aerosol and the second three describe sea-salt aerosol.

6 Running the model

After the code has been compiled (see Section 4), the procedure for running the model is to type:

1. Edit the file `namelist.in` to configure the model.
2. Type `./run.sh` at the command line, followed by `enter` to run the model.

Table 1 describes the parameters in `namelist.in` that can be modified, although obviously you can hack any part of the code if you want to. Note that if you change any Fortran code you will need to type `make` again to compile the code to the machine executable.

7 Experiments

Keep a practical record. Save each figure and make brief notes as you go. Record the parameter values used for each simulation and what changed in the output. You may be asked to interpret these experiments in the assessment.

7.1 Temperature lapse rate of the parcel

The first simulation to do is a standard simulation where we use the background aerosol with the parcel starting at the ground and rising through the atmosphere at 0.3 m s^{-1} to a height of 1200 m. This is the default case, so we just need to type:

```
./run.sh
```

at the command line, followed by `enter`. The code will take a short while to run and an output file will be created. Check that the file is there by typing `ls /tmp/<username>/` where `<username>` is your username.

Exercise: To plot the results of this simulation you will need to run a Python script. Type the following, pressing `enter` after each line

```
python3 python/example_plot_bmm2.py
```

after a short while Python will put a file in the `ls /tmp/<username>/` directory. To download it, go to your SFTP window and type:

```
get /tmp/<username>/vertical.png
```

followed by `enter`, where `<username>` is your username. This should be a plot of the the properties of the parcel in the vertical. It will be discussed in the practical.

Exercise: From the top-right plot, estimate how rapidly temperature decreases with height up to about 800 m altitude. This lapse rate is a characteristic property of dry air rising approximately adiabatically.

Exercise: What happens to the rate of cooling above 800m? Why?

Make a note of the mixing ratio of liquid water in the cloud at the top. This is the mass of liquid water in the parcel of air. Also make a note of the approximate height at which cloud droplets first form.

7.2 Effect of vertical wind-speed

Vertical velocity controls how rapidly the parcel cools with time and how quickly supersaturation develops, which can strongly affect droplet activation. Change `winit` (line 18 in the current `namelist.in`).

Do simulations for $10\times$ less than the default value and $10\times$ more than the default. i.e. set `winit=0.03` and `winit=3.0`.

You should type:

```
nano -l namelist.in
```

then make the change, followed by `Ctrl-X` to exit, then type `Y` followed by `Enter`.

Run the model by typing:

```
./run.sh
```

each time you run the simulation, follow the steps in Section 7.1 to plot the model results and download to your computer.

Exercise: How does the lapse rate look between 0 and 800m on these new simulations?

Exercise: Look at both the liquid water mixing ratio (top left) and the CDNC (Cloud Drop Number Concentration) for these plots. What effect does changing the vertical wind speed have? Why?

Exercise: Approximately what height does the cloud start to form in these simulations? Why?

You should be in the `bin-microphysics-model` directory.

7.3 Increasing humidity (water vapour)

Let's return to the initial default state of the model by typing:

```
git reset --hard
```

Find `rhinit` (line 28 in the current `namelist.in`) by typing:

```
nano -l namelist.in
```

and changing to `rhinit=0.95` and exit / save the file in the usual way.

Run the model and generate the output file and plot in the usual way. Download it using SFTP.

Exercise: How did increasing the initial humidity change the result? Why?

7.4 Adding sea-salt aerosol

From the main model directory `bin-microphysics-model` return to the default:

```
git reset --hard
```

We will now investigate three properties of an added sea-salt mode: number concentration, distribution width and particle size. The simulations are cumulative: do *not* reset the model between runs 1–3.

Run	Sea-salt number	Median diameter	$\ln \sigma_g$
Default	0	100 nm	0.5
1	$1000 \times 10^6 \text{ kg}^{-1}$	100 nm	0.5
2	$1000 \times 10^6 \text{ kg}^{-1}$	100 nm	0.01
3	$1000 \times 10^6 \text{ kg}^{-1}$	800 nm	0.01

1. **Add sea-salt aerosol.** Change the third sea-salt number concentration in `n_aer1` (line 162 in the current file) from zero to $1000 \times 10^6 \text{ kg}^{-1}$:

```
0e6, 0e6, 1000.e6,
```

Keep the default third-mode diameter (`100.e-9`) and width (`0.5`). Run the simulation, make the plot and save the image.

2. **Narrow the sea-salt distribution.** Keep the sea-salt number concentration at `1000.e6` and the diameter at `100.e-9`. Change only the third sea-salt value of `sig_aer1` (line 168) from `0.5` to `0.01`:

```
0.3, 0.3, 0.01,
```

Run the simulation, make the plot and save the image.

3. **Increase the sea-salt particle size.** Keep the sea-salt number concentration at `1000.e6` and `sig_aer1` at `0.01`. Change only the third sea-salt value of `d_aer1` (line 165) from `100.e-9` to `800.e-9`:

```
1e-6, 1e-6, 800.e-9,
```

Run the simulation, make the plot and save the image.

Exercise: Compare run 1 with the default simulation in Section 7.1. What is the effect of adding $1000 \times 10^6 \text{ kg}^{-1}$ sea-salt particles?

Exercise: Compare run 2 with run 1. What is the effect of narrowing the sea-salt size distribution while keeping its number concentration and median diameter fixed?

Exercise: Compare run 3 with run 2. What is the effect of increasing the sea-salt median diameter from 100 nm to 800 nm while keeping its number concentration and distribution width fixed?

7.5 Bonus: Ice Formation

From the main model directory `bin-microphysics-model` return to the default: `git reset --hard`.

Find `z_ctop` (line 98 in the current `namelist.in`) and change the cloud-top height to 12000 m (`z_ctop=12000,`). Do the simulation and the plot. We will discuss the output in the practical.